Vedanshi Mesvaniya

AI Engineer

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Career Objective

- Junior AI Engineer with hands-on experience in Python, LLMs and Retrieval-Augmented Generation (RAG)
- Built AI applications that answer questions from documents and databases, including a company assistant and a personal finance app with multi-LLM fallback
- Skilled in building backend APIs with FastAPI and connecting them to AI features
- Data analysis background with SQL, Power BI and Excel
- Looking for an AI Engineer role to build reliable, real-world AI solutions

Skills

- **Programming & Tools:** Python, SQL, Excel, FastAPI, REST APIs, Git
- **Data Analysis:** Pandas, NumPy, Data Cleaning, Exploratory Data Analysis (EDA)
- **Visualization:** Power BI, Matplotlib, Excel Charts
- **AI & Machine Learning:** LLMs, RAG, Text-to-SQL, Prompt Engineering, Semantic Search, FAISS, Qdrant, Sentence Transformers, LightGBM, Multi-LLM API Integration
- **Databases & Backend:** PostgreSQL, SQLite, SQL dialects, database connectors
- **Frontend (basic):** React
- **Concepts:** Statistics, Hypothesis Testing, Data Interpretation
- **Soft Skills:** Problem Solving, Analytical Thinking, Communication

Projects

Sales Performance Dashboard

Tools: Excel, SQL, Power BI

- Cleaned and analysed 10,000+ retail sales records using Excel and Python to ensure data accuracy and consistency.
- Performed SQL-based analysis to calculate key business metrics including total revenue, average order value, and region-wise performance.
- Identified top-performing products and high-revenue regions to support business insights.
- Built interactive Power BI dashboards to visualize monthly trends, regional sales, and product performance.
- Highlighted 12–15% revenue growth during peak months, enabling better sales planning and decision-making.

AI Chatbot for VuMaxpro

Technologies: Python, LightGBM, FAISS, Sentence Transformers, Pandas, NumPy, Scikit-learn, RAG, PDFPlumber

- Contributed to the development of an AI-powered chatbot for geological and drilling information retrieval.
- Worked on RAG pipelines including document ingestion, semantic search, vector indexing, and knowledge retrieval from technical documents.
- Developed PDF and table extraction workflows for geological and drilling reports.
- Implemented semantic retrieval using FAISS and Sentence Transformers embeddings.
- Built LightGBM models for ROP prediction and drilling parameter optimization.

Stash: AI Personal Finance App

Technologies: Python, FastAPI, React, SQLAlchemy, PostgreSQL, SQLite, LLM APIs

- Built a full-stack AI finance app that converts natural-language inputs like “spent 50 on groceries yesterday” into structured income and expense entries.
- Developed a FastAPI backend with password hashing, signed session cookies, email verification, and strict per-user data isolation.
- Created a 6,000+ item glossary covering English, Hinglish, and Hindi merchants/items, with a learning system that remembers user-defined categories.
- Implemented multi-LLM fallback across 3 providers, retry queues, and offline support to prevent transaction loss during API or network failures.
- Built monthly reports, spending trends, budgets, recurring transactions, and CSV/Excel/PDF export functionality; deployed the application with PostgreSQL.

Antarkosh

Technologies: Python, FastAPI, React, Qdrant, sqlglot, LLMs, RAG, SQL

- Developed a hybrid AI assistant that answers questions across enterprise documents and relational databases through a unified natural-language interface.
- Built a multi-stage document ingestion pipeline for PDF and office files, including OCR, layout-aware and table extraction, embeddings, hybrid dense/sparse retrieval, and reranking.
- Developed a Text-to-SQL pipeline with semantic intent extraction, schema retrieval, business glossary and join context to generate database-aware read queries.
- Implemented AST-based SQL validation and read-only execution safeguards, with query repair and result formatting to handle invalid or unsafe generated queries.
- Added database connector and dialect handling for SQLite, MySQL, PostgreSQL, Microsoft SQL Server, and Oracle; SQLAlchemy-based unified connectivity is planned, not yet implemented.

Experience

Etech Energy Consultancy Pvt. Ltd AI/ML Intern 01/2026 - 05/2026

- Worked on building an AI chatbot using Python and LLMs.
- Used RAG to fetch answers from a custom knowledge base.
- Integrated a prediction API to handle user queries in natural language.
- Assisted in creating a basic intent detection system for user queries.
- Improved chatbot responses by refining prompts and retrieval process.

Param Software AI & Python Automation 06/2025 - To Present

- Developing Antarkosh, an AI assistant that combines enterprise document retrieval with natural-language access to relational business data.
- Building and improving the RAG pipeline with multiformat ingestion, OCR and table extraction, Qdrant hybrid search, embeddings, and reranking.
- Developing Text-to-SQL workflows for schema-aware query generation, dialect-specific SQL handling, and AST-based read-only query validation.
- Debugging and refining SQL + RAG retrieval for large schemas, ambiguous business terminology, query repair, and more grounded responses.
- Extending database connectivity across SQLite, MySQL, PostgreSQL, SQL Server, and Oracle; evaluating SQLAlchemy as a future unified connectivity layer.

Education

Master of Computer Applications (MCA)
Atmiya University, Rajkot | Graduated: 2026 | First Distinction | Aggregate: 91.72%

Bachelor of Computer Applications (BCA)
Saurashtra University, Rajkot | Graduated: 2024 | First Distinction | Aggregate: 72.63%

Certification

- Microsoft Excel Associate (2025)
- Docker 101 – Colaculam (2024)